Exercise Sheet 8

The notation is that of Lecture 8: $\mathfrak{g}$ is semisimple over $\mathbb{C}$, $\mathfrak{h}$ is a Cartan subalgebra, R is the set of roots, h_α is the coroot of α and $\langle \lambda, \alpha^\vee \rangle = \lambda(h_\alpha)$. For $\mathfrak{sp}_4$ we use the form J and the diagonal subalgebra $\mathfrak{h} = \{\mathrm{diag}(a_1, a_2, -a_1, -a_2)\}$ of the lecture. Exercises 8.1–8.3 are matrix, string and coroot calculations. Exercises 8.4–8.6 reconstruct and extend the proofs.

Exercise 8.1 ($\mathfrak{sp}_4$ is semisimple). Let $\mathfrak{g} = \mathfrak{sp}_4$. For each root γ let x_γ be the matrix below.

$$\begin{aligned} x_{\varepsilon_1-\varepsilon_2} &= E_{12} - E_{43}, & x_{-\varepsilon_1+\varepsilon_2} &= E_{21} - E_{34}, \\ x_{\varepsilon_1+\varepsilon_2} &= E_{14} + E_{23}, & x_{-\varepsilon_1-\varepsilon_2} &= E_{41} + E_{32}, \\ x_{2\varepsilon_1} &= E_{13}, & x_{-2\varepsilon_1} &= E_{31}, \\ x_{2\varepsilon_2} &= E_{24}, & x_{-2\varepsilon_2} &= E_{42}. \end{aligned}$$

For each root γ put $k_\gamma = [x_\gamma, x_{-\gamma}]$.

(i) Show that $x_\gamma \in \mathfrak{g}$, that $[h, x_\gamma] = \gamma(h) x_\gamma$ for $h \in \mathfrak{h}$, and that $\mathfrak{h}$ and the eight x_γ form a basis of $\mathfrak{g}$.

(ii) Prove that $\mathfrak{g}$ has no nonzero abelian ideal. Deduce that $\mathfrak{g}$ is semisimple.

(iii) Prove $C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$, and that $\mathfrak{h}$ is a maximal toral subalgebra.

Exercise 8.2 (Strings in $\mathfrak{sp}_4$). The roots of $\mathfrak{sp}_4$ are $\pm\varepsilon_1 \pm \varepsilon_2$ and $\pm 2\varepsilon_i$. For each pair, find the α-string through β, the integers r, q and $\langle \beta, \alpha^\vee \rangle$. Check the last value against the elements k_γ of Exercise 8.1.

(a) $\alpha = \varepsilon_1 - \varepsilon_2$, $\beta = \varepsilon_1 + \varepsilon_2$.

(b) $\alpha = 2\varepsilon_1$, $\beta = \varepsilon_1 - \varepsilon_2$.

(c) $\alpha = 2\varepsilon_2$, $\beta = 2\varepsilon_1$.

(d) $\alpha = \varepsilon_1 + \varepsilon_2$, $\beta = 2\varepsilon_1$.

In (d), compute $[x_{-\alpha}, x_\beta]$ and identify its root space. Compare (a) with the α_1-string through $\alpha_2 = 2\varepsilon_2$ in the lecture. Finally, put $x = x_{\varepsilon_1-\varepsilon_2}$ and $y = x_{-\varepsilon_1+\varepsilon_2}$. Compute $\mathrm{ad}\, x$ and $\mathrm{ad}\, y$ along the string $2\varepsilon_2, \varepsilon_1+\varepsilon_2, 2\varepsilon_1$, and match the coefficients with the basis $v_j = f^j v_0$ of V_2.

Exercise 8.3 (Lengths and coroots of C_2, and a warning). Put $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = 2\varepsilon_2$ in $\mathfrak{sp}_4$.

(i) Use the trace formula $\kappa(h, h') = \sum_{\gamma \in R} \gamma(h)\gamma(h')$ of the lecture to compute κ on $\mathfrak{h}$. Deduce the form on $\mathfrak{h}^*$, find (α_1, α_1) and (α_2, α_2), and rescale so that long roots have squared length 2. Lecture 16 normalises invariant forms in this way.

(ii) Identify the real diagonal matrices in $\mathfrak{h}$ with $\mathbb{R}^2$ by $\mathrm{diag}(a_1, a_2, -a_1, -a_2) \mapsto (a_1, a_2)$. Write the coroots of $\varepsilon_1 \pm \varepsilon_2$ and $2\varepsilon_i$ as vectors. Show that long roots have short coroots for the dot product, and explain why the dot product is the right comparison.

(iii) Show that $\varepsilon_1 + \varepsilon_2 = \alpha_1 + \alpha_2$, but $h_{\varepsilon_1+\varepsilon_2} \neq h_1 + h_2$.

(iv) Compute $b_{ij} = 2(h_j \cdot h_i)/(h_i \cdot h_i)$ and compare it with the Cartan matrix (a_{ij}) of the lecture.

(v) A source using the transpose convention $a'_{ij} = \alpha_i(h_j)$ prints

$$\begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$$

for this pair. Which simple root would a reader using the course convention take to be long?

Exercise 8.4 (Root triples are rigid). Let $\alpha \in R$ and let $(x_\alpha, h_\alpha, y_\alpha)$ be a root triple.

(i) Show that $h_{-\alpha} = -h_\alpha$, and that $(y_\alpha, -h_\alpha, x_\alpha)$ is a root triple for $-\alpha$.

(ii) Show that the root triples for α are exactly $(cx_\alpha, h_\alpha, c^{-1}y_\alpha)$ with $c \in \mathbb{C}^\times$, and that $\kappa(x_\alpha, y_\alpha) = 2/(\alpha,\alpha)$ for every root triple.

(iii) For $\mathfrak{g} = \mathfrak{sl}_2$ with $\mathfrak{h} = \mathbb{C}h$ and $\alpha(h) = 2$, use $\kappa(h,h) = 8$ from Lecture 4 to find t_α, (α,α) and h_α. Evaluate $\kappa(ih, ih)$.

Exercise 8.5 ($\mathfrak{sl}_3$ under a root $\mathfrak{sl}_2$). Let $\alpha = \varepsilon_1 - \varepsilon_2$ in $\mathfrak{sl}_3$, with root triple $(E_{12}, E_{11} - E_{22}, E_{21})$. Decompose $\mathfrak{sl}_3$ under $\operatorname{ad} S_\alpha$ into irreducible submodules, giving a highest vector for each. Check the number of summands against (W2).

Exercise 8.6 (Bounds on Cartan integers and strings). Let $\alpha, \beta \in R$ with $\beta \neq \pm\alpha$.

(i) Prove that $\langle \beta, \alpha^\vee \rangle \langle \alpha, \beta^\vee \rangle \in \{0, 1, 2, 3\}$.

(ii) Prove that the two integers are both zero or have the same sign.

(iii) Deduce that the α-string through β has at most four roots.